Steps Towards Migraine Prediction

Predicting Migraine through Digital Diaries and Step and Stair Count

Jocelyn A. Barsky-Moore, BScH[1], Christopher L. Metts, MD[3], Mara Getz, MA, MPH[1], Megan Lacritz, MA[1], Rebecca E. Klapper, MA, MEd, CMPC[1], Kensei I. Maeda, MA[1], Elizabeth K. Seng, PhD[1,23]

[1]Ferkauf Graduate School of Psychology, Yeshiva University, Bronx, NY [2]Saul R. Korey Department of Neurology, Albert Einstein College of Medicine, Bronx, NY [3]Medical University of South Carolina, Charleston, SC

Background

Migraine medications are most effective early in an attack, therefore attack prediction would allow for more effective treatment.

While wearable sensor models show promise in attack prediction, they are not accessible to everyone. iPhones allow for active data collection from surveys in addition to passively count **steps and stairs.**

Avoidance of physical activity like stairs is a diagnostic feature of migraine (ICHD-3, the international diagnostic criteria), but the behaviour is not commonly studied or included in attack prediction.

Daily phone diaries have reached an AUC of .83 for next day attacks (Tsai et al., 2025, a machine learning study of digital headache diaries).

AIM

Evaluate both active and passive data as predictors of headache day.

Methods

Participants completed daily headache diaries for 30 days, while passive iPhone data was collected.

1. **Eligibility:** at least 21 diary days, at least 3 headache days, and Apple Health step and stair data.
2. **Diary:** headache status, medication, and features; on headache days. Sleep, pain, symptoms, acute medication, interference, the Allodynia Symptom Checklist, and premonitory and neurologic symptom checklist; everyday.
3. **Models:** L2-penalized logistic regression and random forest, fit separately for each participant using prior-day features to predict next-day headache. Class imbalance was handled with class weights. Performance was summarized with balanced accuracy, F1, and accuracy (not AUC). Analyses used Python (scikit-learn); code was written with Cursor and reviewed by the first author.
4. **Evaluation:** rolling or holdout validation, one step ahead fallback when data were sparse. Balanced accuracy used to choose best model.

Participants

37 eligible → **26** modeled

Participants were people who reported features of episodic migraine. Modeled participants had at least 20 usable next day rows, both headache and headache free outcomes, and enough non missing features.

Results

0.71 mean balanced accuracy

5 participants at 1.0

6 at or below chance

Median balanced accuracy 0.71; mean F1 macro 0.64; mean accuracy 0.71.

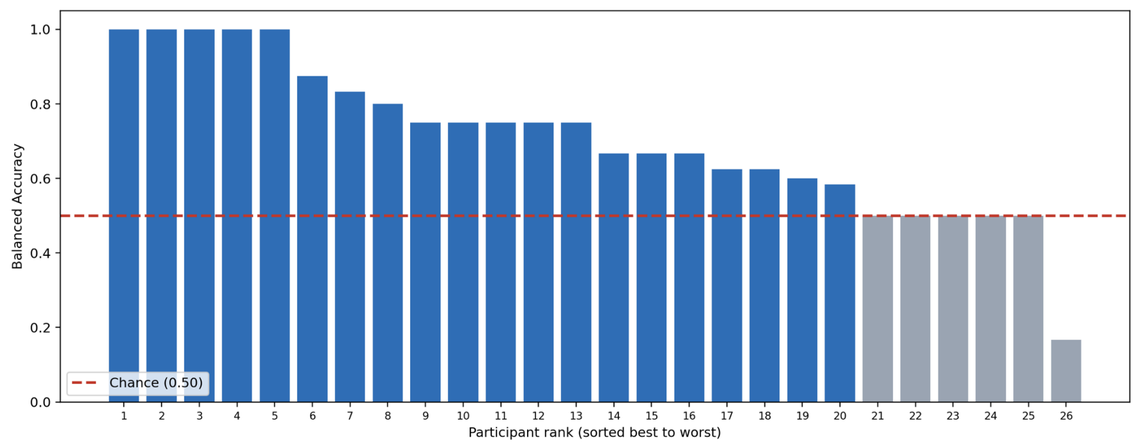


Figure 1. Balanced Accuracy by Participant Best Model (n= 26).

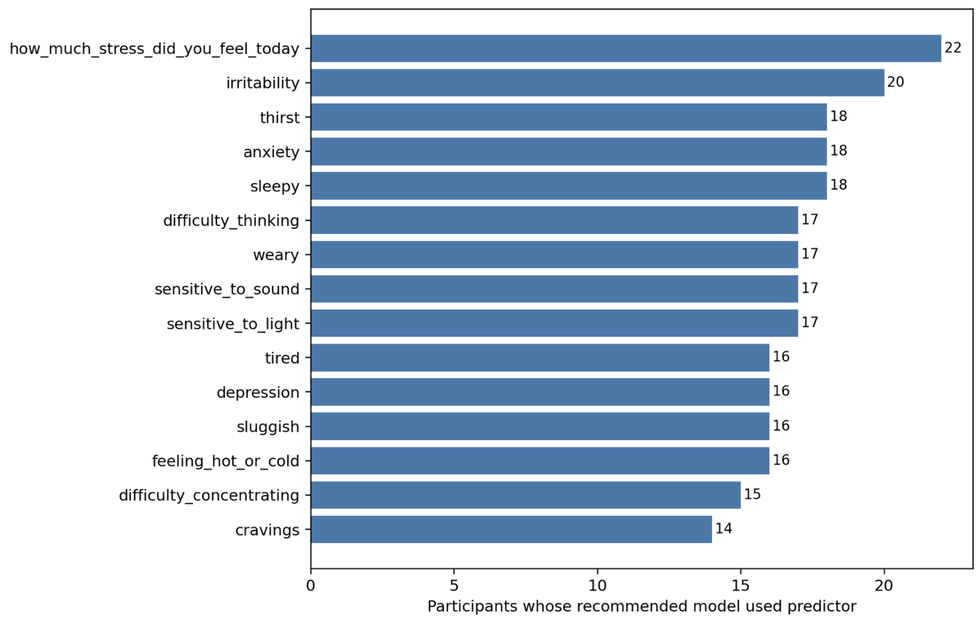


Figure 2. Top Predictors in Final Recommended Models (n=26)



Figure 3. Best model type for participants (n=26)

Most frequent day before predictors

DIARY
- Stress (22)
- Irritability (20)
- Thirst (18)
- Anxiety (18)
- Sleepiness (18)

PASSIVE
- Steps (4)
- Walking speed (4)
- Step length (4)
- Basal energy burned (4)
- Distance walking and running (4)

Direction and size of effects varied between participants.

Conclusions

1. Step and stair count may help predict migraine.
2. Prediction needs at least 3 headache days and minimal missing data.
3. People may need different model types, feature sets, and number of features.
   - Cannot use base model

Next steps

1. These are pilot results; models were likely overfitting.
   - Collect more than 30 days and more headache days per person.
2. Study other passive data that may be used for prediction.
3. Building and studying use of predictor profiles

References

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